

Student:

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Email:

Time on Task:

1 hour, 16 minutes

Progress:

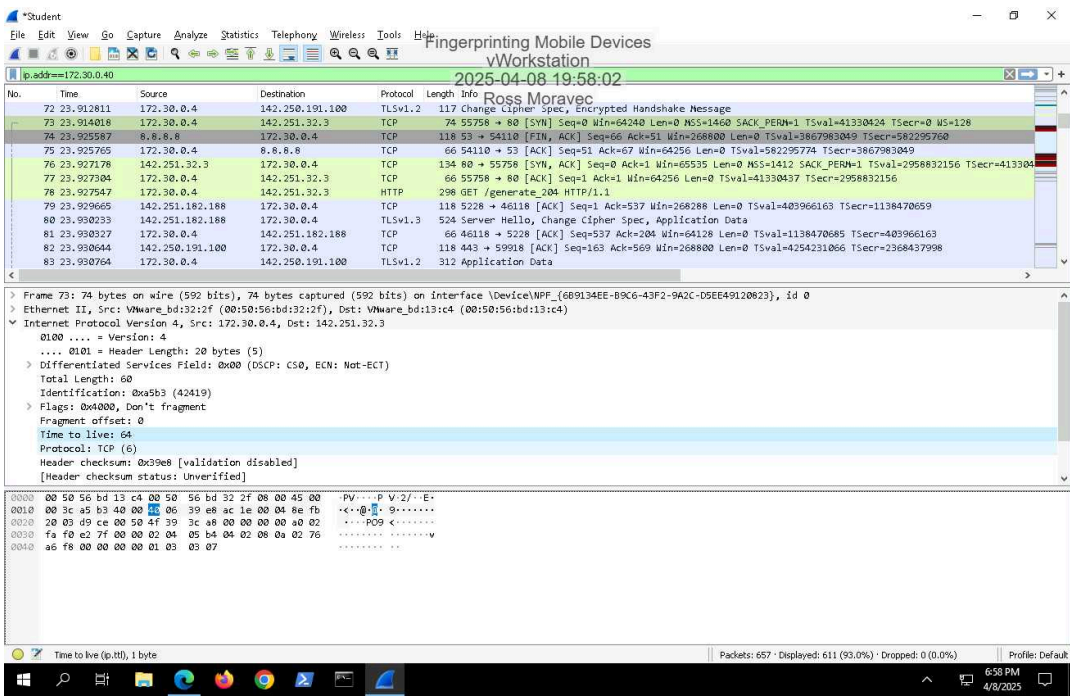
100%

Report Generated: Tuesday, April 8, 2025 at 9:59 PM

Section 1: Hands-On Demonstration

Part 1: Perform Passive Fingerprinting with Wireshark

14. Make a screen capture showing the Time to live field.



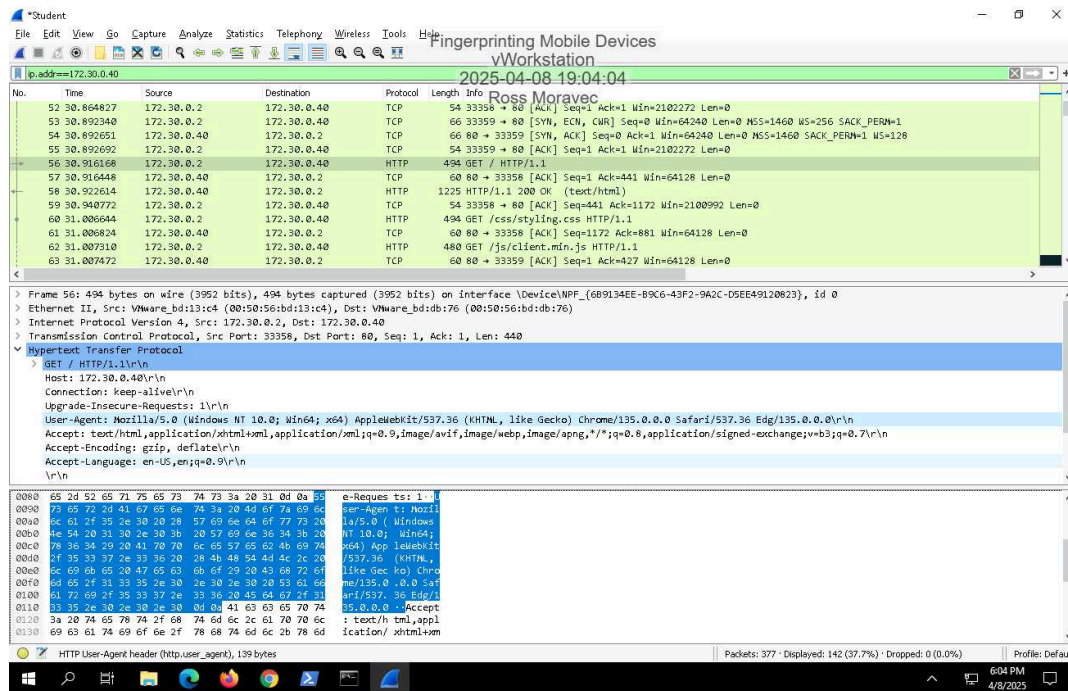
18. Make a screen capture showing the User-Agent string for the Chrome browser session.

Wireshark packet capture showing a Chrome browser session. The packet list shows a GET request to http://172.30.0.40. The packet details pane shows the User-Agent string: Mozilla/5.0 (Linux; Android 9; VMware Virtual Platform) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/92.0.4515.166 Safari/537.36. The packet bytes pane shows the raw data of the request.

23. Make a screen capture showing the User-Agent string for the Firefox browser session.

Wireshark packet capture showing a Firefox browser session. The packet list shows a GET request to http://172.30.0.40. The packet details pane shows the User-Agent string: Mozilla/5.0 (Android 9; Mobile; rv:91.0) Gecko/91.0 Firefox/91.0. The packet bytes pane shows the raw data of the request.

29. Make a screen capture showing the User-Agent string for the Edge browser session from the vWorkstation.

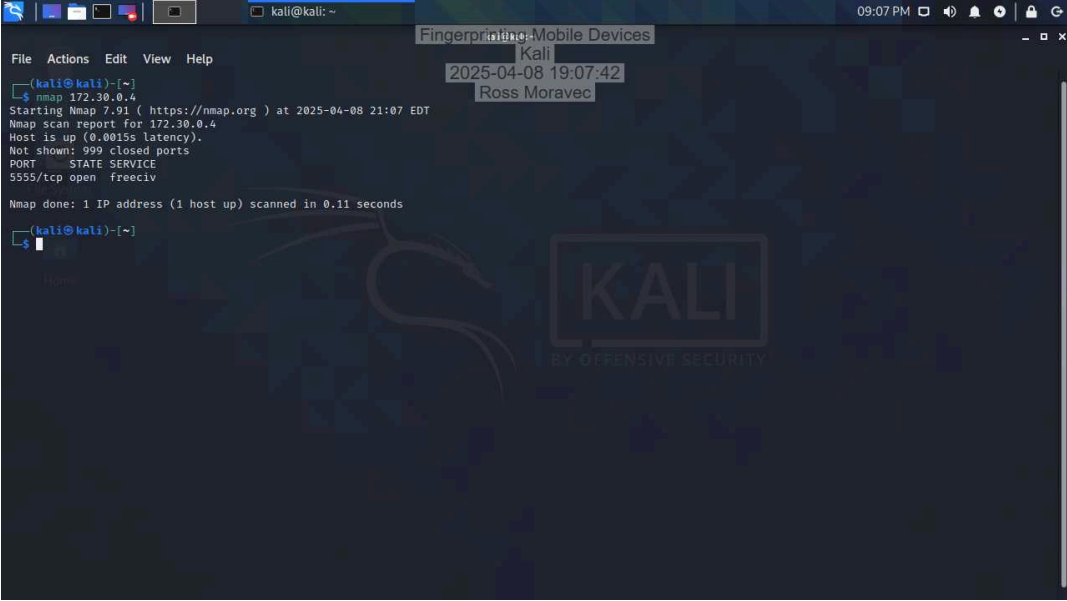


30. Compare the three User-Agent strings. How do they differ? How are they similar?

All three User-Agent strings begin with the common “Mozilla/5.0” token but differ primarily by operating system (e.g., “Android 9” vs. “Windows NT 10.0; Win64; x64”), browser engine (e.g., “AppleWebKit” for Chrome and Edge vs. “Gecko” for Firefox), and device type (“Mobile” for Android, “Win64; x64” for Windows). Chrome on Android references “Mobile Safari/537.36” alongside “Chrome/[version],” Firefox on Android uses “Gecko/[version]” and “Firefox/[version],” and Edge on Windows includes both “Chrome/[version]” and “Edg/[version],” reflecting its Chromium base. Despite these variations, they all follow a standardized structure starting with “Mozilla/5.0,” then specifying OS, engine, and browser versions.

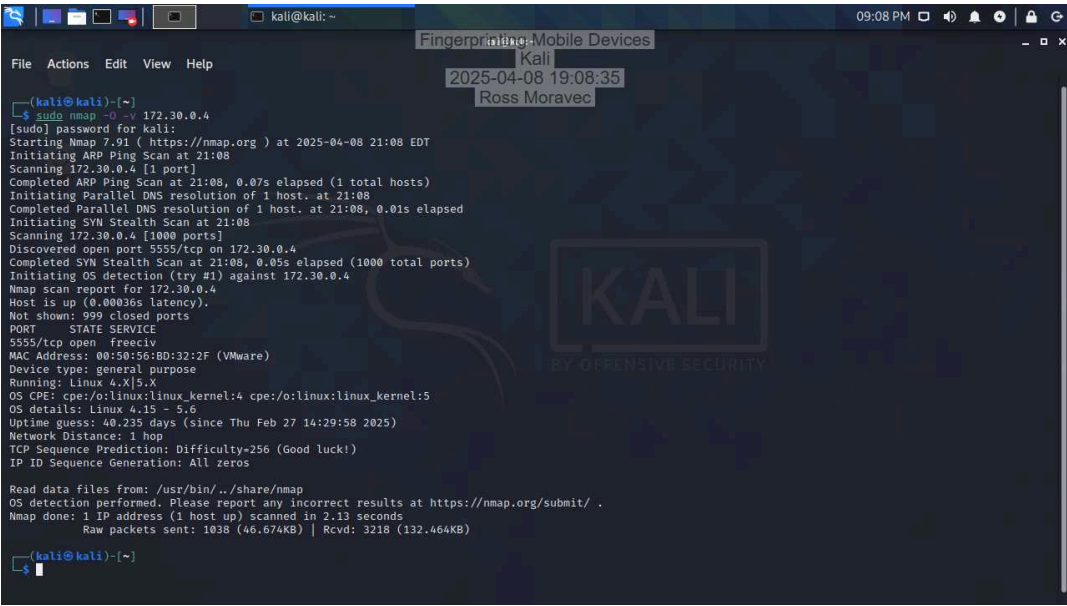
Part 2: Perform Active Fingerprinting with Nmap

5. Make a screen capture showing the results of the default Nmap scan.



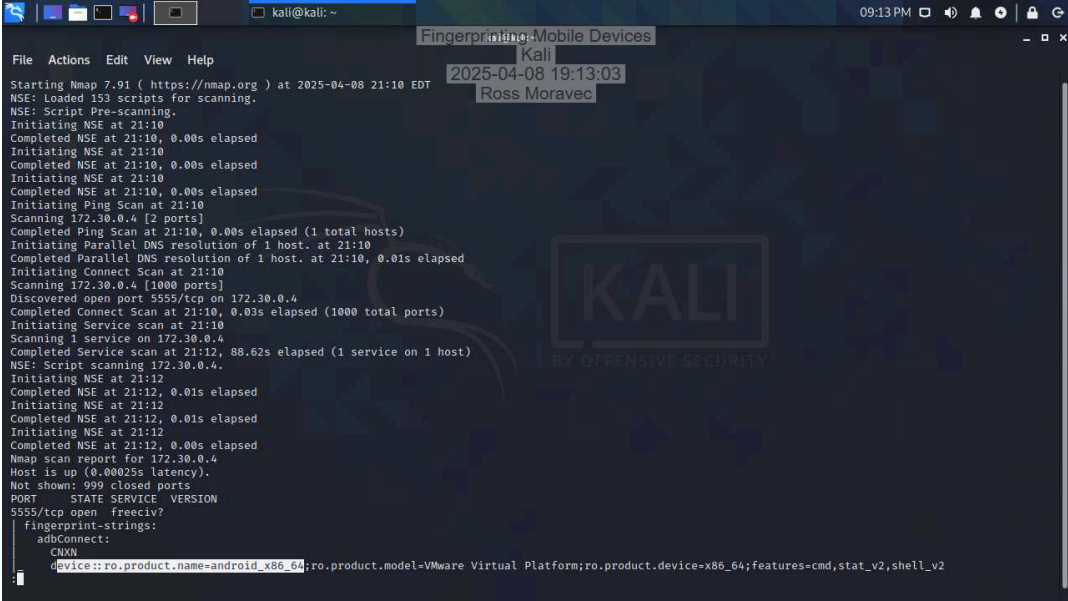
```
kali@kali: ~  
File Actions Edit View Help  
[kali@kali]~  
$ nmap 172.30.0.4  
Starting Nmap 7.91 ( https://nmap.org ) at 2025-04-08 21:07 EDT  
Nmap scan report for 172.30.0.4  
Host is up (0.0015s latency).  
Not shown: 999 closed ports  
PORT      STATE SERVICE  
5555/tcp  open  freeciv  
Nmap done: 1 IP address (1 host up) scanned in 0.11 seconds  
[kali@kali]~  
$
```

7. Make a screen capture showing the results of the -O -v Nmap scan.



```
kali@kali: ~  
File Actions Edit View Help  
[kali@kali]~  
$ sudo nmap -O -v 172.30.0.4  
[sudo] password for kali:  
Starting Nmap 7.91 ( https://nmap.org ) at 2025-04-08 21:08 EDT  
Initiating ARP Ping Scan at 21:08  
Scanning 172.30.0.4 [1 port]  
Completed ARP Ping Scan at 21:08, 0.07s elapsed (1 total hosts)  
Initiating Parallel DNS resolution of 1 host. at 21:08  
Completed Parallel DNS resolution of 1 host. at 21:08, 0.01s elapsed  
Initiating SYN Stealth Scan at 21:08  
Scanning 172.30.0.4 [1000 ports]  
Discovered open port 5555/tcp on 172.30.0.4  
Completed SYN Stealth Scan at 21:08, 0.05s elapsed (1000 total ports)  
Initiating OS detection (try #1) against 172.30.0.4  
Nmap scan report for 172.30.0.4  
Host is up (0.00036s latency).  
Not shown: 999 closed ports  
PORT      STATE SERVICE  
5555/tcp  open  freeciv  
MAC Address: 00:50:56:BD:32:2F (VMware)  
Device type: general purpose  
Running: Linux 4.X|5.X  
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5  
OS details: Linux 4.15 - 5.6  
Uptime guess: 40.235 days (since Thu Feb 27 14:29:58 2025)  
Network Distance: 1 hop  
TCP Sequence Predictions: Difficulty=256 (Good luck!)  
IP ID Sequence Generation: All zeros  
Read data files from: /usr/bin/./share/nmap  
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 2.13 seconds  
Raw packets sent: 1038 (46.674KB) | Rcvd: 3218 (132.464KB)  
[kali@kali]~  
$
```

11. Make a screen capture showing the product name in the Nmap output.

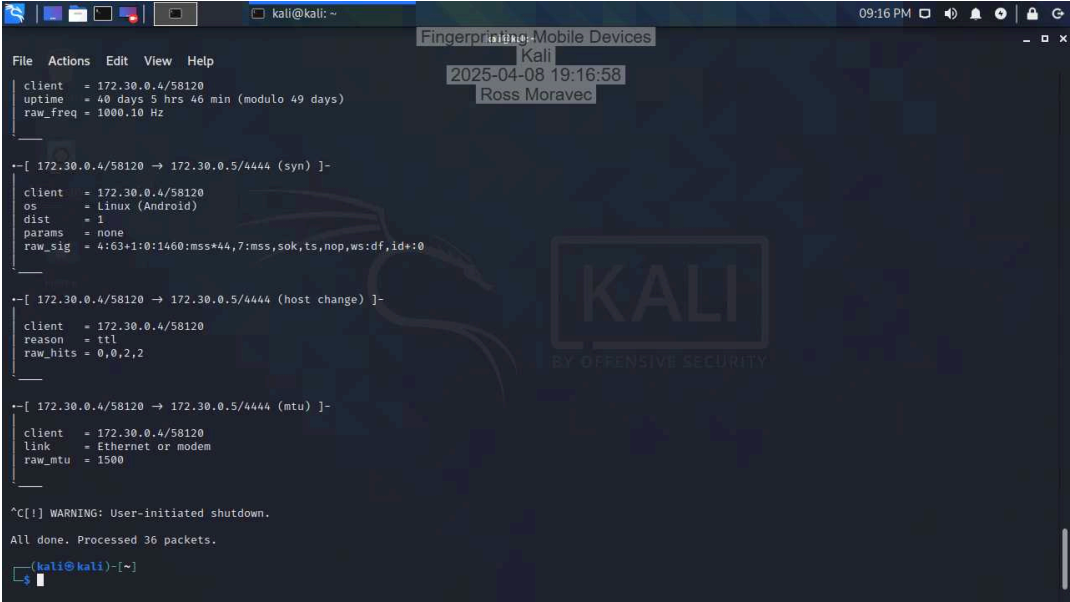


```
File Actions Edit View Help
Starting Nmap 7.91 ( https://nmap.org ) at 2025-04-08 21:10 EDT
NSE: Loaded 159 scripts for scanning.
NSE: Script Pre-scanning.
Initiating NSE at 21:10
Completed NSE at 21:10, 0.00s elapsed
Initiating NSE at 21:10
Completed NSE at 21:10, 0.00s elapsed
Initiating NSE at 21:10
Completed NSE at 21:10, 0.00s elapsed
Initiating Ping Scan at 21:10
Scanning 172.30.0.4 [2 ports]
Completed Ping Scan at 21:10, 0.00s elapsed (1 total hosts)
Initiating Parallel DNS resolution of 1 host. at 21:10
Completed Parallel DNS resolution of 1 host. at 21:10, 0.01s elapsed
Initiating Connect Scan at 21:10
Scanning 172.30.0.4 [1000 ports]
Discovered open port 5555/tcp on 172.30.0.4
Completed Connect Scan at 21:10, 0.03s elapsed (1000 total ports)
Initiating Service scan at 21:10
Scanning 1 service on 172.30.0.4
Completed Service scan at 21:12, 88.62s elapsed (1 service on 1 host)
NSE: Script scanning 172.30.0.4.
Initiating NSE at 21:12
Completed NSE at 21:12, 0.01s elapsed
Initiating NSE at 21:12
Completed NSE at 21:12, 0.01s elapsed
Initiating NSE at 21:12
Completed NSE at 21:12, 0.00s elapsed
Nmap scan report for 172.30.0.4
Host is up (0.00025s latency).
Not shown: 999 closed ports
PORT      STATE SERVICE VERSION
5555/tcp  open  freeciv?
Fingerprint-strings:
  adbConnect:
    CNXX
  device::ro.product.name=android_x86_64;ro.product.model=VMware Virtual Platform;ro.product.device=x86_64;features=cmd,stat_v2,shell_v2
```

Section 2: Applied Learning

Part 1: Perform Passive Fingerprinting with p0f

7. Make a screen capture showing the p0f result identifying 172.30.0.4 as Linux (Android).



```
File Actions Edit View Help
client = 172.30.0.4/58120
uptime = 40 days 5 hrs 46 min (modulo 49 days)
raw_freq = 1000.10 Hz

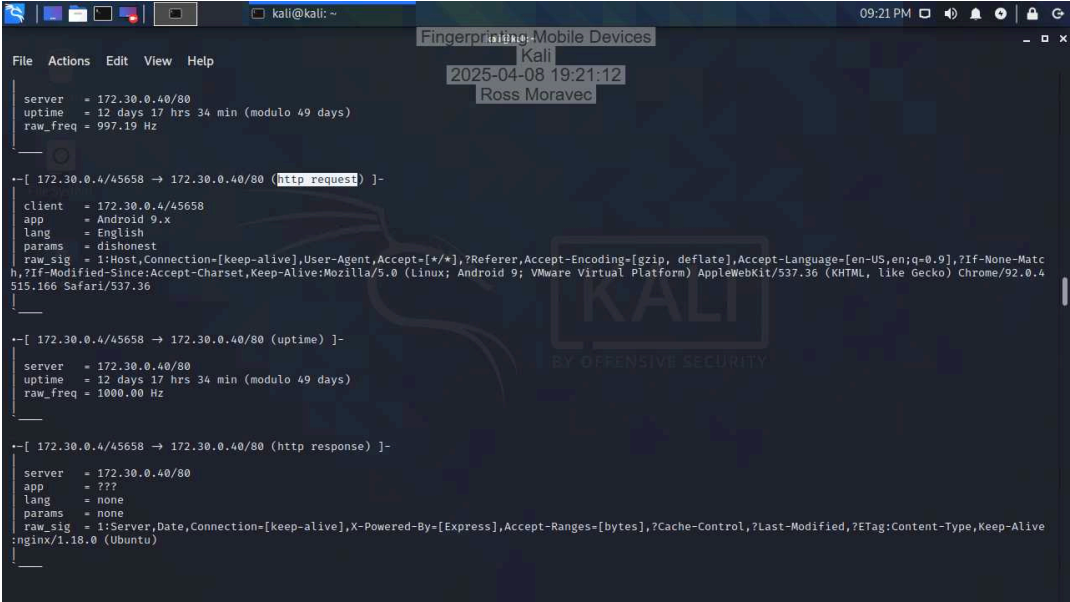
--[ 172.30.0.4/58120 → 172.30.0.5/4444 (syn) ]-
client = 172.30.0.4/58120
os = Linux (Android)
dist = 1
params = none
raw_sig = 4:63+1:0:1460:mss*44,7:mss,sok,ts,nop,ws:df,id+:0

--[ 172.30.0.4/58120 → 172.30.0.5/4444 (host change) ]-
client = 172.30.0.4/58120
reason = ttl
raw_hits = 0,0,2,2

--[ 172.30.0.4/58120 → 172.30.0.5/4444 (mtu) ]-
client = 172.30.0.4/58120
link = Ethernet or modem
raw_mtu = 1500

^C[!] WARNING: User-initiated shutdown.
All done. Processed 36 packets.
(kali@kali)-[~]
```

14. Make a screen capture showing the p0f result identifying 172.30.0.4 as Android 9.x.



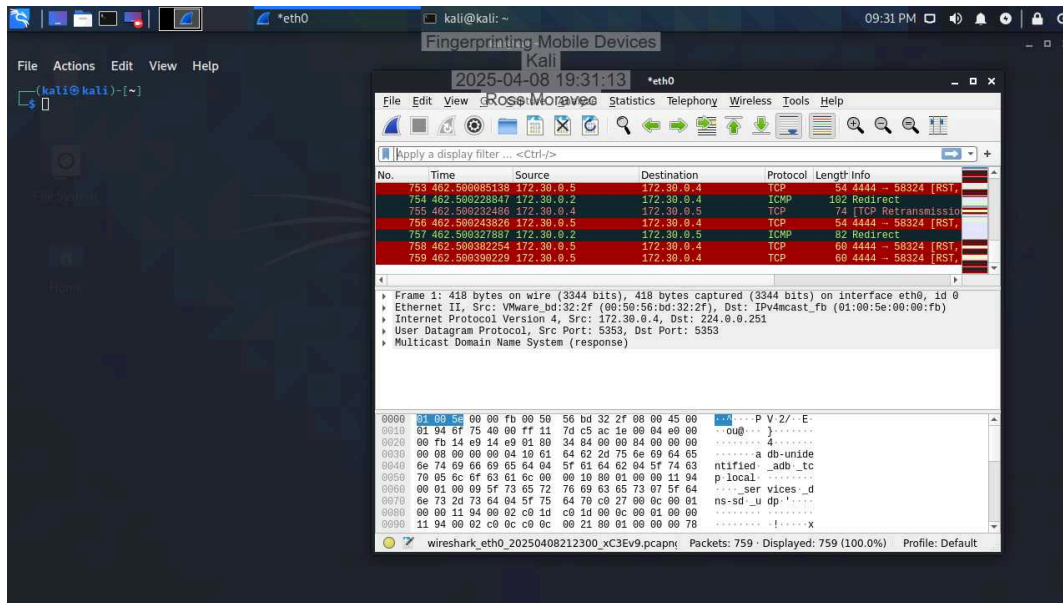
```
File Actions Edit View Help
server = 172.30.0.40/80
uptime = 12 days 17 hrs 34 min (modulo 49 days)
raw_freq = 997.19 Hz

--[ 172.30.0.4/45658 → 172.30.0.40/80 (http request) ]-
client = 172.30.0.4/45658
app = Android 9.x
lang = English
params = dishonest
raw_sig = 1:Host,Connection=[keep-alive],User-Agent,Accept=[/*/*],?Referer,Accept-Encoding=[gzip, deflate],Accept-Language=[en-US,en;q=0.9],?If-None-Matc
h,?If-Modified-Since:Accept-Charset,Keep-Alive:Mozilla/5.0 (Linux; Android 9; VMware Virtual Platform) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/92.0.4
515.166 Safari/537.36

--[ 172.30.0.4/45658 → 172.30.0.40/80 (uptime) ]-
server = 172.30.0.40/80
uptime = 12 days 17 hrs 34 min (modulo 49 days)
raw_freq = 1000.00 Hz

--[ 172.30.0.4/45658 → 172.30.0.40/80 (http response) ]-
server = 172.30.0.40/80
app = ???
lang = none
params = none
raw_sig = 1:Server,Date,Connection=[keep-alive],X-Powered-By=[Express],Accept-Ranges=[bytes],?Cache-Control,?Last-Modified,?ETag:Content-Type,Keep-Alive
:nginx/1.18.0 (Ubuntu)
```

19. Make a screen capture showing the traffic generated by Nmap in Wireshark.

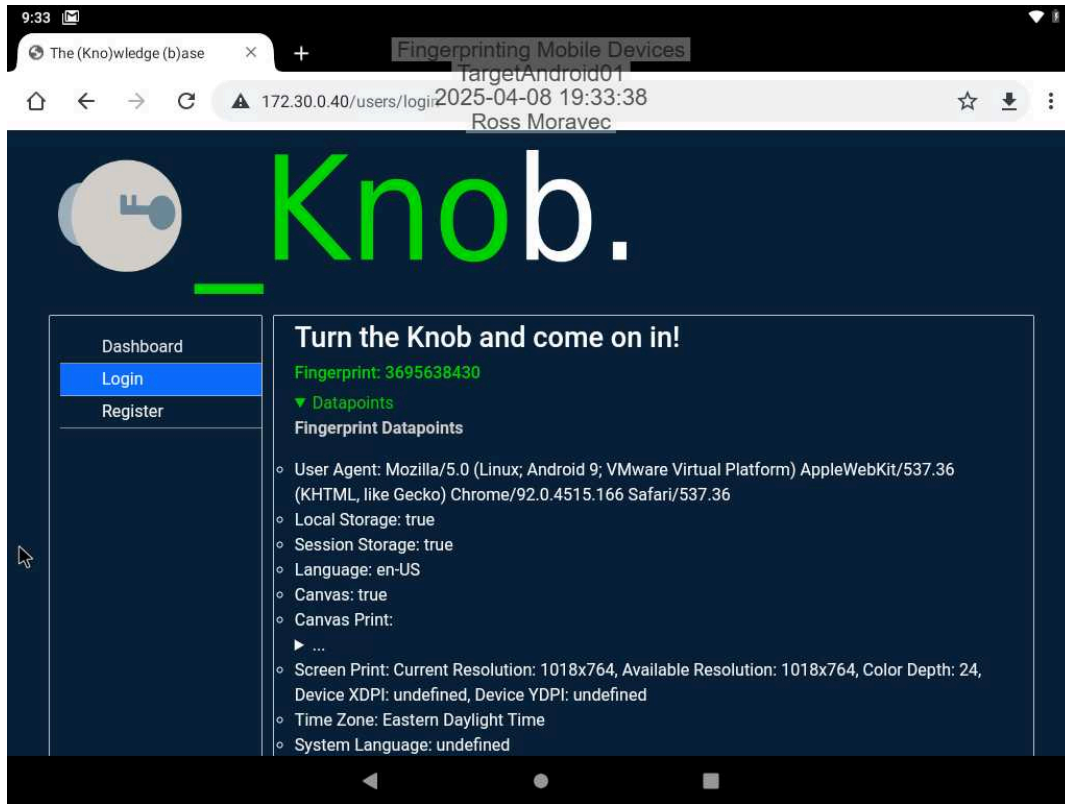


24. Compare the Wireshark results from the p0f scan to the Nmap scan.

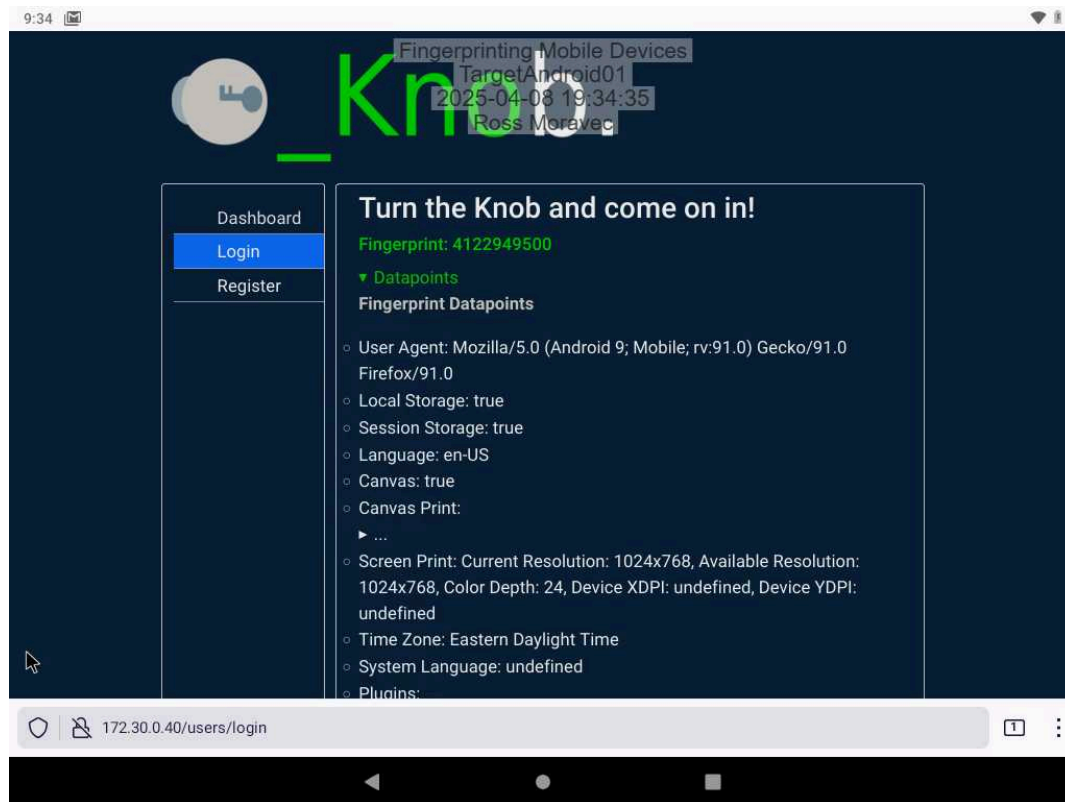
When p0f is running, Wireshark shows only the normal TCP traffic that occurs as the Android device initiates or responds to connections (e.g., HTTP GET requests). Since p0f uses passive fingerprinting, it does not send additional probes, resulting in minimal overhead or suspicious packets in Wireshark. In contrast, when Nmap is scanning the same device, Wireshark captures a flood of probe and response packets, often targeting multiple ports and using various scan techniques (e.g., SYN, SYN-ACK, or OS-detection packets). This extra traffic from Nmap's active approach stands out in Wireshark, potentially triggering alarms on firewalls or intrusion detection systems, whereas p0f's passive method blends in with ordinary network flows.

Part 2: Perform Active Fingerprinting with ClientJS

4. Make a screen capture showing the **Chrome fingerprint on TargetAndroid01**.



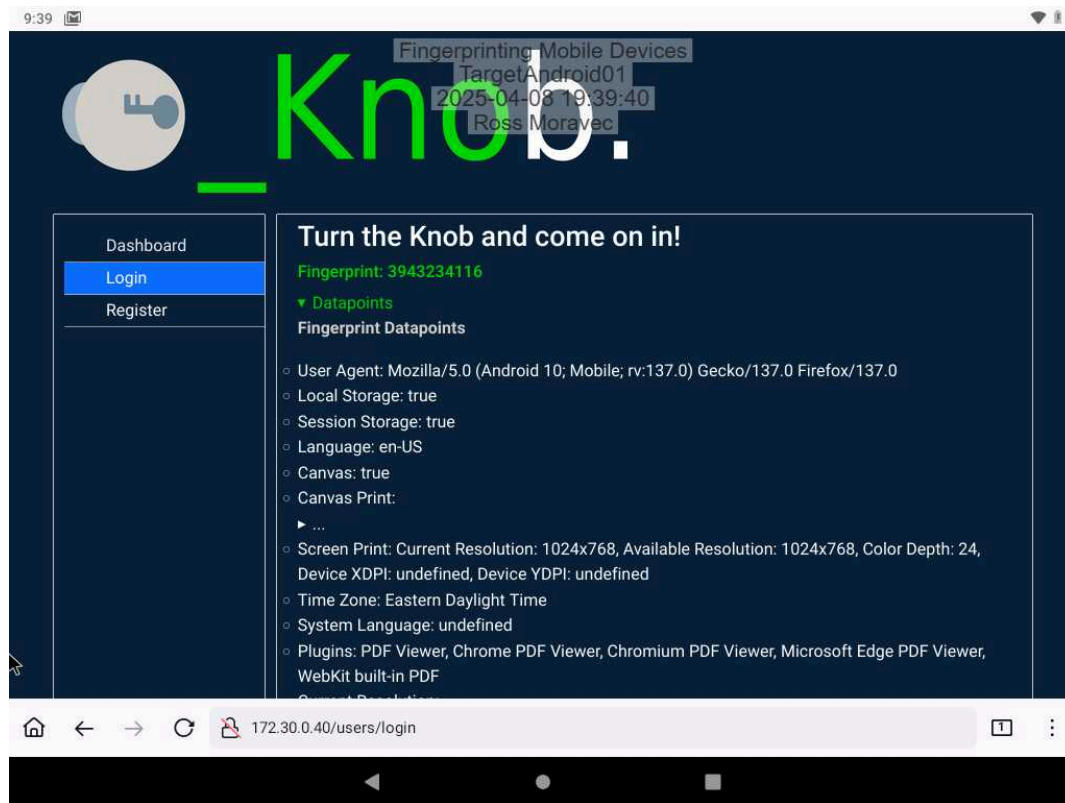
9. Make a screen capture showing the Firefox fingerprint on TargetAndroid01.



10. Identify the differences between the Datapoints values generated for Chrome and the Datapoints values generated for Firefox.

When comparing Chrome and Firefox on the same Android device, the main differences lie in each browser's user agent, reported screen resolution, and rendering engine details. Chrome's User-Agent string includes "Chrome/[version]" and "AppleWebKit/537.36," whereas Firefox reports "Gecko/[version]" and "Firefox/[version]." These variances impact the resulting "Fingerprint" hash and may slightly alter certain reported datapoints like screen resolution or plugin detection. However, both browsers share core attributes—such as the same OS version (Android 9), local/session storage availability, default language settings (e.g., en-US), and time zone—which ensures some similarity in their overall datapoint profiles.

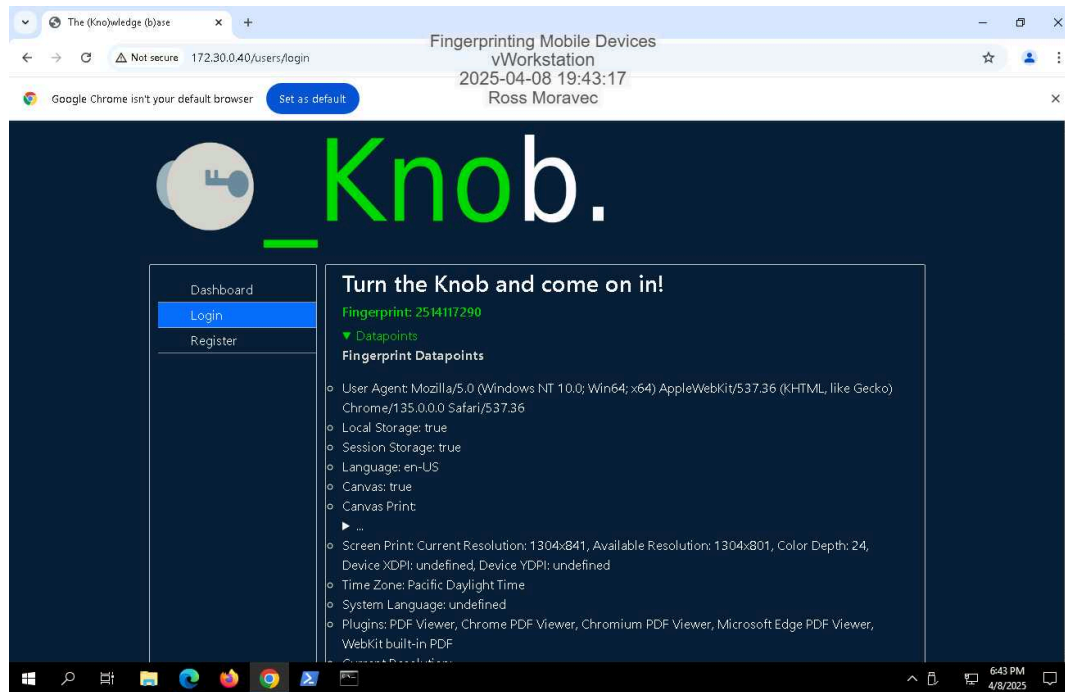
16. **Make a screen capture** showing the **updated Firefox fingerprint on TargetAndroid01**.



17. **Identify** the differences between the Datapoints values for Firefox before and after the update.

After updating Firefox, the User-Agent and associated version details changed from something like "Mozilla/5.0 (Android 9; Mobile; rv:91.0) Gecko/91.0 Firefox/91.0" to a newer string (e.g., indicating Android 10 and Firefox 137). This resulted in a different fingerprint hash and may also have updated certain datapoints such as available plug-ins, the screen resolution print, and system language

20. **Make a screen capture** showing the **Chrome fingerprint on the vWorkstation**.



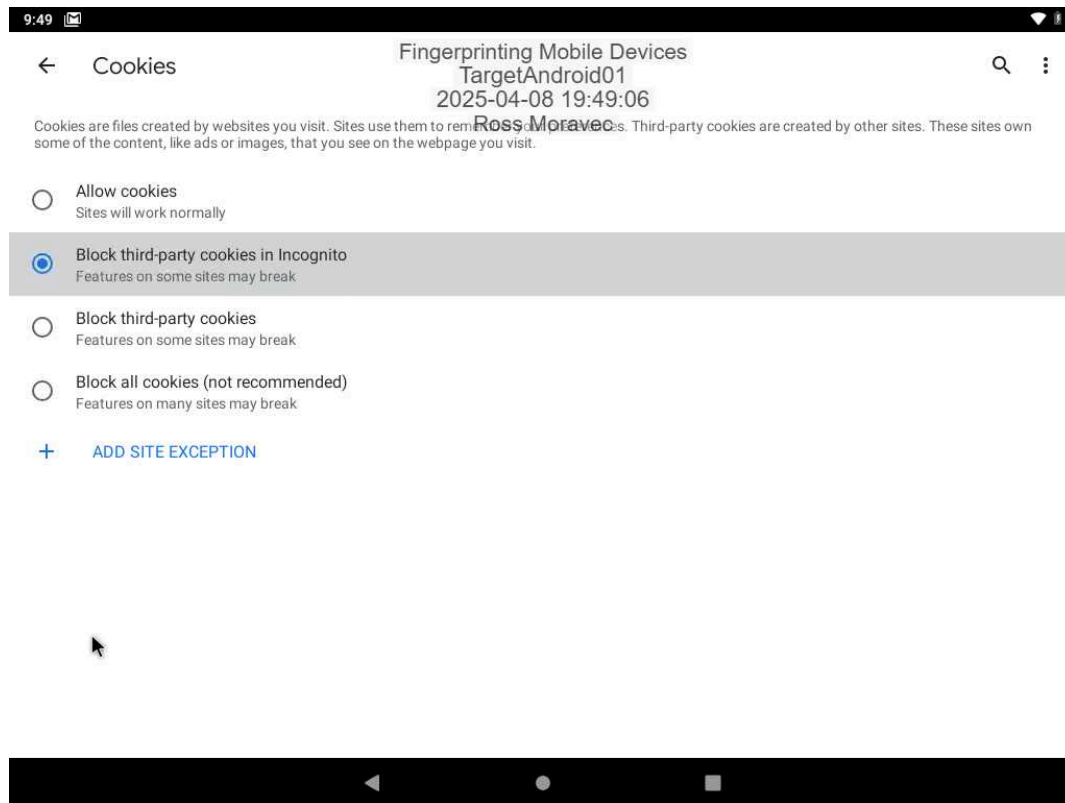
21. **Identify** the differences between the Datapoints values on the vWorkstation and TargetAndroid01.

On the vWorkstation, the User-Agent identifies Windows (“Windows NT 10.0; Win64; x64”) and Chrome with a desktop form factor, while on TargetAndroid01 it references Android 9 (or 10) plus a mobile environment. This difference leads to distinct screen resolution reports—Windows typically shows a higher or differently scaled resolution, while Android resolutions might be smaller

Section 3: Challenge and Analysis

Part 1: Disable Third-Party Cookies

Make a screen capture showing the **cookie settings** in **Chrome** on **TargetAndroid01**.



What would happen if you switched to a different option for allowing versus blocking cookies, and how would it hinder attempts to track your activities online? Would there be any drawbacks?

If you switch from allowing all cookies to blocking some or all cookies—especially third-party cookies—you reduce the ability of advertisers and analytics services to track your browsing across multiple sites. This helps protect your privacy by limiting the data collected about your online activities. However, there can be drawbacks: certain websites may rely on third-party cookies for embedded content or single sign-on features, which can break site functionality or force you to log in more frequently.

Part 2: Test Your Browser's Tracking Protection

Make a screen capture showing the **Cover Your Tracks** assessment results.

